Hasan Mohd Hussain

Machine Learning Engineer | Generative AI, LLMs, RAG & Agentic AI
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PROFESSIONAL SUMMARY

Machine Learning Engineer with a **Master's in Computer Engineering**, specializing in **Generative AI**, **LLMs**, **RAG** and **agentic AI** systems. Experienced in building, evaluating and deploying ML applications with **Python**, **FastAPI**, **Docker** and **Azure OpenAI**.

EDUCATION

The University of Texas at Dallas — **Richardson, TX**
Master of Science (MS) in Computer Engineering | GPA **3.74 / 4.0** — Aug 2024 – Aug 2026
Relevant Coursework - Machine Learning, NLP, Data Structures & Algorithms, Database Design, Computer Architecture, Embedded Systems

University of Mumbai — **Mumbai, India**
Bachelor of Engineering (B.E.) in Computer Engineering | GPA **7.88 / 10.0** — Aug 2017 – Jul 2021

SKILLS

AI / ML: Machine Learning, Deep Learning, NLP, Computer Vision, CNNs, Transformers, Supervised Learning, Fine-Tuning, Hyperparameter Tuning, Model Evaluation
Generative AI & LLMs: LLMs, Generative AI, RAG, Agentic AI, AI Agents, Multi-Agent Systems, Prompt Engineering, Embeddings, Semantic Search, LLM Evaluation, LLM-as-a-Judge, Knowledge Graphs
LLM Frameworks & APIs: LangChain, LangGraph, OpenAI API, Azure OpenAI, Gemini API, Hugging Face, Vector Databases
ML Libraries: PyTorch, Scikit-learn, Hugging Face Transformers, NumPy, Pandas, Matplotlib, Seaborn
MLOps & Deployment: Model Deployment, Model Serving, FastAPI, Flask, REST APIs, Docker, Kubernetes, CI/CD, RabbitMQ
Cloud Platforms: AWS, GCP, Azure
Databases: PostgreSQL, MySQL, MongoDB, SQLite, ChromaDB, RDF4J, SPARQL
Programming Languages: Python, SQL, C++, C#, JavaScript
Tools & Practices: Git, GitHub, GitLab, Linux, Shell Scripting, PyTest, Playwright, Claude Code, Data Structures & Algorithms, OOP, Agile/Scrum

WORK EXPERIENCE

Mphasis — **New York, United States**
Machine Learning Engineer — May 2025 – Aug 2025, Jun 2026 – Aug 2026

- Developed an autonomous **LLM-based QA agent** (**Python, Flask, Playwright**) that browser-tests a production mortgage-income dashboard end-to-end and grades results against ground-truth keys via a **multi-provider LLM layer** (Claude, GPT-4o, Gemini, self-hosted Qwen).
- Applied **LLM-as-a-judge evaluation** to replace manual QA, cutting per-case test time by **95%** (minutes to seconds).
- Built a **synthetic data generator** for **5 financial document types** (W-2, paystub, VOE, Schedule C, P&L) with computed answer keys, scaling **evaluation coverage to 200 cases per run** with zero manual authoring.
- Built a **RAG** system ingesting **PDFs** into **ChromaDB** with a **FastAPI** dashboard and **Azure OpenAI** LLM, improving retrieval by **30%**.
- Optimized **Python backend logic** for full-stack AI apps (**FastAPI, LangChain, AWS**), improving **API response performance by 20%**.

Dot9 Games — **Mumbai, India**
Game Developer Intern — Apr 2024 – Aug 2024

- Built gameplay systems in **Unity** and **C#** from game design documents, shipping mechanics that raised **user engagement by 25%**.
- Collaborated with developers, artists and designers in **Agile/Scrum** sprints, using **Git** for version control and iterative delivery.

Shree Infotech — **Mumbai, India**
Software Engineer — Apr 2023 – Mar 2024

- Developed **Python/Flask backend services** and internal tools that automated manual team workflows and reduced operational overhead.
- Containerized applications with **Docker** to standardize development and deployment environments across the team.

Nilee Games — **Mumbai, India**
Game Developer Intern — Nov 2022 – Mar 2023

- Built a **2D physics-based driving game** in **Unity/C#** within **two weeks**, contributing to **500K downloads** in its first month.
- Increased **user engagement by 30%** by tuning gameplay mechanics: terrain design, vehicle upgrades and a progression system.

TECHNICAL PROJECTS

RAG Document QA System with Gemini Integration

- Implemented a **Retrieval-Augmented Generation (RAG)** pipeline with **LangChain**, **ChromaDB** and **Hugging Face embeddings** for semantic search, document chunking and PDF text retrieval.
- Integrated the **Google Gemini** LLM to generate context-aware answers, improving query accuracy by **35%**.

LangGraph Agentic Nutrition System

- Built a **multi-agent** nutrition AI system (**LangGraph, GPT-4o-mini**) with **ReAct agents** and a critic-revise loop using score-based routing.
- Streamed live Think-Act-Observe traces per agent via **Flask SSE**, generating a 7-day meal plan, supplement protocol and safety-audited report.

Deep Learning Image Classification (MLP vs CNN) – PyTorch

- Developed **MLP** and **CNN** architectures in **PyTorch** on MNIST and CIFAR-10 using SGD/Adam, BatchNorm and Dropout.
- Ran **30+ hyperparameter experiments** (learning rate, batch size, regularization): **99%+ accuracy** on MNIST, **78%** on CIFAR-10.

CERTIFICATIONS

Anthropic: Claude Code 101, Claude 101, Introduction to Claude Cowork | **DeepLearning.AI:** Supervised Machine Learning: Regression and Classification, AI For Everyone
IBM: Machine Learning with Python, Building AI Agents and Agentic Workflows | **Udemy:** The Complete SQL Bootcamp